

MECHATRONICS

1 Definition

Mechatronics is the **integration** of mechanical engineering, electronics, control systems and computer technology to design and develop intelligent systems that can perform tasks automatically, accurately and efficiently.

In Simple Words

Mechatronics means combining machines with electronics, control and computers to make smart systems that can **think, sense, decide** and **work automatically**.

2 Main Components of Mechatronics

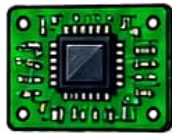
Mechanical Systems

- Structure
- Design
- Moving parts



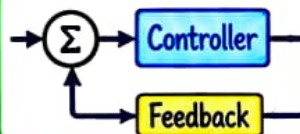
Electronics

- Sensors
- Signal conditioning
- Microcontrollers



Control Systems

- Feedback control
- Decision making (PID, etc.)



Computer Systems

- Programming
- Embedded systems
- Data processing



MECHATRONICS

Integration of all these to create intelligent and automated systems

3 Examples of Mechatronic Systems

1. Industrial Robot Arm



- Used in car manufacturing for welding, painting etc.
- High precision and repeatability.

2. CNC Machine



- Computer controlled machining.
- New design can be made just by changing code (G-code).

3. ABS in Automobile



- Sensors detect wheel speed.
- Prevents wheel locking during emergency braking.

4. DSLR Camera



- Auto-focus mechanism controlled by electronics and software.
- Compact and smart design.

5. Automatic Washing Machine



- Uses sensors, controller and motor for automatic operation.
- Stops and shows error if any fault occurs.

6. Drone (UAV)



- Uses sensors and control algorithms.
- Stable flight and automatic operation.

7. Elevator System



- Automatic control for smooth and safe movement.
- Uses sensors and controller.

8. Smart Home System



- Integrates sensors, controllers and communication.
- Automatic control and energy saving.

HISTORY OF MECHATRONICS

WORD ORIGIN



**Tetsuro Mori
(1969)**

(Yaskawa Electric,
Japan)

- Coined the term "**Mechatronics**" in 1969 to describe a new field that integrates mechanical engineering, electronics and control systems.



YASKAWA

Yaskawa Electric
Corporation
(Japan)

Mecha- + **-tronics**
(Mechanical) (Electronics)

Mechatronics

Combination of Mechanical
and Electronics with Control

TIMELINE (1969 - PRESENT)

1969

Word Coined

- Tetsuro Mori coined the term "Mechatronics".
- Integration of mechanics, electronics and control.



1971

Trademark Registered

- Yaskawa Electric Corporation (Japan) registered "Mechatronics" as an official trademark.
- Used for their advanced integrated products.



1982

Released to Public Domain

- Yaskawa renounced the trademark.
- The term became a common word used worldwide.
- Wider use in industry and academia.



1990s

Global Academic Acceptance

- Included in engineering curriculum.
- Universities worldwide (IITs, MIT, Cambridge etc.) started research and development.
- Became a well-established interdisciplinary field.



**2000s
-
Present**

Smart Era / Industry 4.0

- Integration of AI, IoT, Robotics, Big Data and Cloud.
- Smart manufacturing, autonomous systems.
- Cyber-physical systems and digital twins.



CORE EQUATION

Mechanical

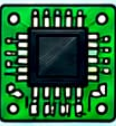
(Design, Structure,
Actuators, Motion)



+

Electronics

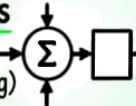
(Sensors, Signal
Conditioning,
Microcontrollers)



+

Control Systems

(Feedback Control,
PID, System Modelling)



+

Computing

(Software, Embedded
Systems, AI, Data
Processing)



=

MECHATRONICS

Evolution of Mechatronics

Future
Smart Era

2000s -
Present

Smart Era / Industry 4.0

- Integration of AI, IoT, Robotics, Big Data, Cloud
- Smart manufacturing, Autonomous systems
- Cyber-physical systems, Digital twins



Growth
&
Global
Expansion

1990s

Global Academic Acceptance

- Mechatronics included in engineering curriculum
- Universities worldwide (IITs, MIT, Cambridge etc.)
- Research and development expanded
- Became a well-established interdisciplinary field



- IITs
- MIT
- Cambridge
- Stanford
- etc.

Global
Recognition

1982

Trademark Renounced / Released to Public Domain

- Yaskawa Electric released the trademark
- The term "Mechatronics" became a common word
- Used globally in industry and academia



- Used worldwide
- Wider adoption
- Innovation in different industries

Foundation
&
Birth

1971

Yaskawa Registered Official Trademark

- Yaskawa Electric Corporation (Japan)
- Registered "Mechatronics" as an official trademark
- Used for their advanced integrated products
- Focus on combination of mechanical and electronic control

YASKAWA
Yaskawa Electric
Corporation
(Japan)



1969

Tetsuro Mori Coined the Term "Mechatronics"

- Tetsuro Mori (Yaskawa Electric, Japan)
- Combined "Mecha" (Mechanics) + "Electronics"
- Described a new field integrating mechanical engineering, electronics and control systems.



Tetsuro Mori
(Japan)

"Mechatronics"
Mecha = Mechanics
+
Electronics

Evolution of Mechatronics



STARTING POINT (Pre-Mechatronics)

Pure Mechanical System

- Only mechanical components
- No electronics or control
- Manual operation

Characteristics

- Mechanical motion
- Simple structure
- Human operation
- No intelligent control

Concept: Integration

Phase 1: Primary Level (1960s - 1970s)



Washing Machine
Timer (Electromechanical)

Key Features:

- Basic integration
- Simple electronics + mechanical units
- Electromechanical systems

Examples:

- Washing machine timer (Electromechanical)
- Automatic door



Automatic Door

Impact

- First integration of electronics with mechanics
- Improved functionality
- Beginnings of automation

Concept: Intelligence

Phase 2: Secondary Level (1980s)



ABS
ABS in Cars

Key Features:

- Microprocessors & Microcontrollers
- Decision making power
- Programmable control

Examples:

- ABS in Cars
- CNC Machines



CNC Machines

Impact

- Intelligent control systems
- Higher accuracy
- Reduced human intervention
- Development of industrial automation

Concept: Networking & Compactness

Phase 3: Tertiary Level (1990s)



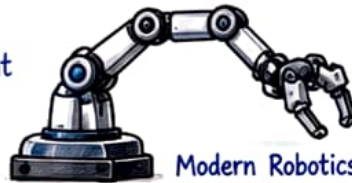
Auto-focus Camera

Key Features:

- Software integration
- Advanced control algorithms
- Smart sensors
- Compact and efficient systems.

Examples:

- Auto-focus Camera
- Modern Robotics



Modern Robotics

Impact

- Real-time control
- More precise and adaptive systems
- Integration of hardware and software
- Compact and smart products

Concept: Smart Era

Phase 4: Quaternary Level (2000s - Present)



Autonomous Vehicles

Key Features:

- IoT & AI integration
- Wireless networks
- Machine learning / Self-learning
- Cyber-physical systems
- Advanced sensors and cloud

Examples:

- Autonomous Vehicles
- Surgical Robots
- Industry 4.0 (Smart Factories)



Surgical Robots



Industry 4.0
(Smart Factories)

Impact

- Smart, connected and autonomous systems
- Higher productivity
- Intelligent decision making
- Creation of smart factories and smart society